

COFFEE PACKAGING QUALITY INSPECTION REPORT

Real-Time AI Seal Inspection and Physical Leak Detection

1. Report Information

Packet ID	PKT-20260901154004-B4A1
Report ID	RPT-20260901-8FC72B
Inspection Date	01 September 2026
Inspection Time	03:45:33 PM
Inspection Module	Packet Seal and Leak Quality Inspection

2. Overall Inspection Summary

Inspection	Result
Real-Time AI Overheat Inspection	OVERHEAT_DETECTED
Physical Packet Leak Test	GOOD
Overall Quality Decision	REJECT

3. Final Quality Decision

REJECT

Potential overheat defect was detected by the real-time AI inspection system.

4. Real-Time Two-Stage AI Inspection

Detected Seal Regions	2
Overheat Detected	YES
Highest Overheat Confidence	63.9%
AI Inspection Status	OVERHEAT_DETECTED

Seal-by-Seal Results

Seal	Seal Confidence	Overheat	Overheat Confidence
Seal 1	78.7%	NO	-
Seal 2	64.7%	NO	-

Real-Time Inspection Evidence



5. Physical Packet Leak Detection

Leak Test Result	GOOD
Initial Value	273729
Average Value	245074.9
Threshold	275000
Minimum	243223
Maximum	246291
Range	3068
Reading Count	10
Final Device Status	TOP_REACHED

Load Cell Readings

Reading Number	Sensor Value
1	243223
2	243899
3	244346
4	244808
5	245046
6	245430
7	245696
8	245918
9	246092
10	246291

6. Recommended Action

- Hold or reject the affected packet.
- Inspect sealing temperature settings.
- Verify sealing dwell time and heater condition.
- Inspect subsequent packets from the same batch.

7. Inspection System Information

AI Stage 1	Seal Region Object Detection
AI Stage 2	Overheat Defect Object Detection

Camera	IP Webcam Real-Time Video
Physical Controller	Arduino Uno
Force Measurement	HX711 Load Cell System
Mechanical Test	Motorized Packet Pressure Mechanism
Safety / Position Control	Top and Bottom Limit Switches

This report was generated automatically using the coffee packet quality inspection system.